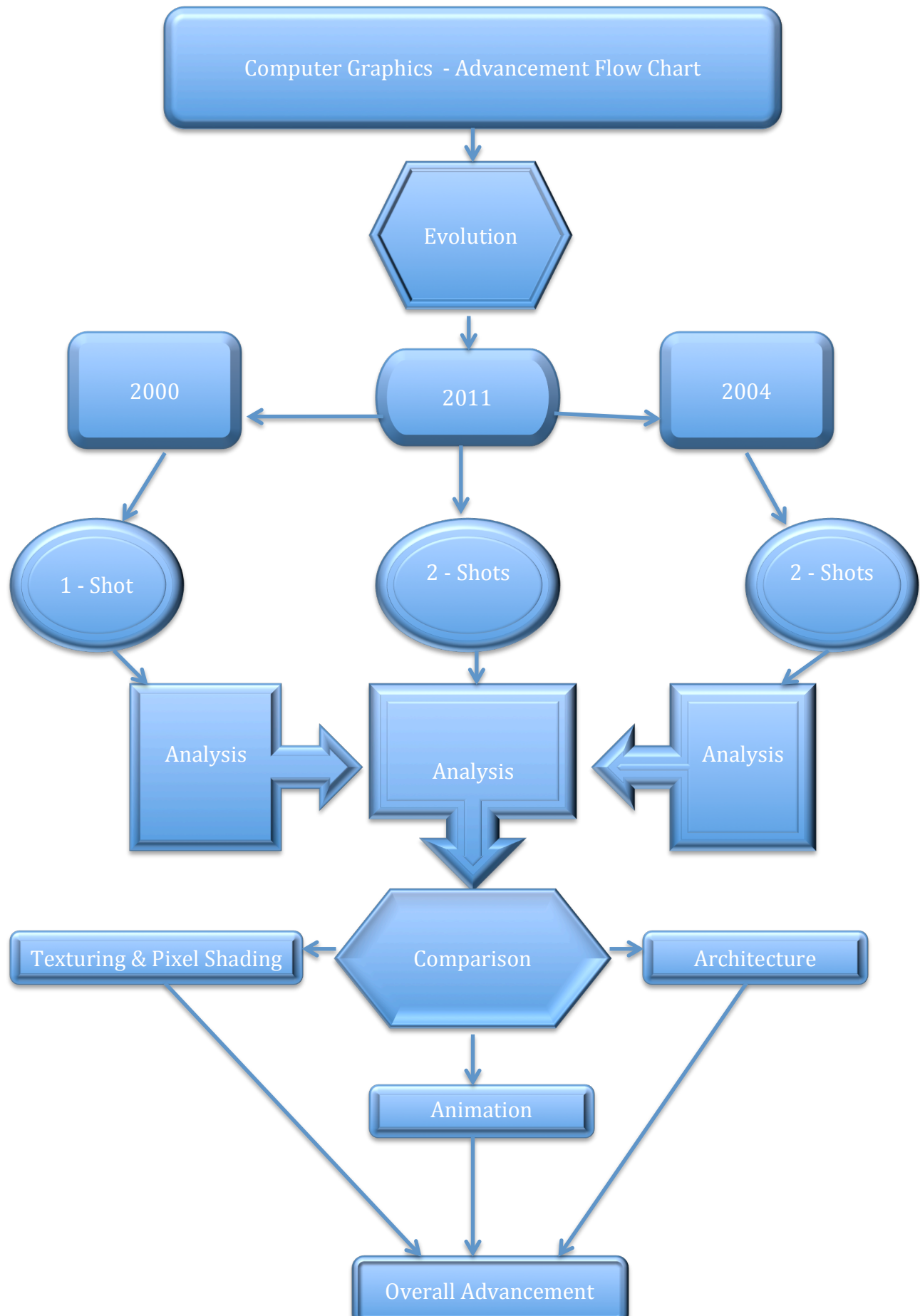




Computer Graphics:

William Fetter coined computer Graphics in 1960, this science describes everything on computers that is not text or sound. This analogy leads us to the study of the following topics for this assignment.

- The manipulation and representation of image data.
- The technology used to create and manipulate images.
- Method used to digitally synthesizing and manipulating visual content.

Computer Graphics Advancement:

The goal of this assignment is to highlight the biggest advancements of the computer graphics from the year 2000 until the year 2011. To be able to detect these advancements, five shots from three different sources will be analyzed and interpreted in a way it gives the reader an understanding of how much and how fast computer graphics has been advancing throughout the years. Moreover, each shot will be analyzed independently and compared to provide the user with a better understanding.

Computer Graphics – 2000 Shot Analysis

Shot Name: Avenue Amy

Director: Joan Raspo

Shot: 1

Graphical Analysis:

The motion of the objects, including people, is slow. The lighting and shading of the objects was not of great definition since human expressions were more based on mouth, eyes, and body motion than lighting effect to reflect the mood and attitude of the person without need of body motion. The color distribution was lacking of tone and proper shading according to the space and position of the object or person, in which they should have been more descriptive. Finally, this shot's architecture was not geographically perceptive in terms of the environments and atmospheres that they tried to illustrate. The reason why the imagery dimension was poorly defined is because the actors were given an illustrated look through a process of subtle image manipulation with blurs to obtain an animated look.

Computer Graphics – 2004 Shot Analysis**Shot Name:** Birthday Boy**Director:** Sejong Park**Shot:** 1**Graphical Analysis:**

The motion of the objects, including the people, dimension and speed are properly distributed and neutralized with the atmosphere that the shot is intended to reflect. The color dissemination is objectively well applied, and it illustrates a good and almost real texture. The whole emphasis is to convey the kid's innocence, and this was well transmitted by simulating good human features, gestures, and face expressions that came to life when right proportion of light and proper shading was applied. However, the lighting position in the apple of the kid got to disconnect the nature and real perception given by the visual effect to this character.

This animated shot was well created in sense of visual effect application, but I personally think that the lighting technique was over used in this shot to a point where the graphics get to lose dimension and texture visibility. Even though the texture and object dimension give a good description in reflecting a war type environment., the architecture still felt more gaming based than real.

Computer Graphics – 2004 Shot Analysis**Shot Name:** BD - Tree**Director:** Dough L. James & Dines K. Pai**Shot:** 2**Graphical Analysis:**

This shot introduces a new way of looking at objects through a computer. It shows how simple is to implement this methods as well as its performance for a variety of real-time and complex off-line deformable simulations. BD - Tree method provides us with a better and faster algorithm that allows us to detect output – sensitive collision for reduced deformable models at similar cost as to standard algorithms for rigid bodies. The object dimension, and the sensitiveness to regions within the same pattern, can be greatly appreciated. Thus, this is a new method to give life to objects as well as to produce more dimensionality in their implementations to make them sensitive to any interaction in the world of computer graphics. This big discover in performing collision detection is done with reduced deformable models at costs comparable to collision detection with rigid objects.

Computer Graphics – 2011 Shot Analysis**Shot Name:** Assassin's Creed Brotherhood E3 Traile**Director:** UBISOFT**Shot:** 1**Graphical Analysis:**

The graphics demonstrated in this shot are astonishingly outstanding to my eyes. The texture looks terrific and its dimensionality is as well constructed as implemented. The color distribution and its depth are well precision in this shot, I was able to notice a high use of object shading, and to my understanding, it actually gives objects a sense of more real and dimensionally reachable to the eyes. It looks obvious that this shot is suing unified and geometry shadier to unify, or complement, the architecture to its colors. This is perfectly distributing all colors into lighting with other visual effects that well coordinates to the body motion as well as to the architecture environment.

Face features, gestures, and expressions are amazingly well perfection, and also illustrated as how a normal human being would look if transported to another dimension. The objects give a sense of dimensional freedom. In fact, this animated shot takes the user's imagination from the real world to the world that the author is relating to.

Computer Graphics – 2011 Shot Analysis**Shot Name:** Sintel**Director:** Colin Levy**Shot:** 2**Graphical Analysis:**

The architecture used to produce different environments enhances with the gestures, expressions and body motion that are dimensionally simulated. Having said that, his shot focuses mostly on expressions, the characters emotions, and the feeling through face illustration and lighting techniques, which were very well applied when it comes to communicating through the eyes.

This author got to transmit the pain, need and other emotions only through the graphics implementations that made the characters gestures, explicitly the eyes, seen just as real as those of human. Lighting was highly used and developed to denote features and objects in every scenario. Even though this animated shot was well built, still its kind of gaming based, real and virtual world mixture did not draw me as much as the characters emotions and gestures simulations. In fact, this animated shot develops a high tree dimensional architecture amazing and well-developed expressions so lighting can be applied to the objects interactions with the environment, place, and different scenarios to which the characters are subjected to.

Comparison

The comparison of the above shots graphical analysis will be mostly based on areas that computer graphics has drastically improved throughout the years, and that have proved the future enhancement between the real world and the virtual world. These are as follows:

Texturing and Pixel Shading:

Early 2000's texture compression form part of the new generation, where shots such as avenue amy demonstrated by giving to real actors a animation look. This was done by very expensive techniques, process, and subtle image manipulation that blur the image. Even though this looks nowadays like a poor technique, this was actually one of the techniques that open the doors to different ways of creating animated characters. Then, in the late 2004's the texture and pixel shading was very advanced. Shot 1 illustrated how fast the texture and pixel shading progressed just in four years. Texture, color distribution, object lighting and shading gave a huge turn to computer graphics. Shot 2 provided with object sensitivity to texture and pixel shading, this made computer graphics development much easier to implement and develop to project dimensional objects with the proper color and texture distribution. Finally, in late 2011's a whole new world in texturing and pixel shading has come. The vertex transform and lighting, along with a unified geometry shading, gave life to objects make its texture, including people, to look just as real as humans. Moreover, shot 2 shows a great improvement in lighting and object shading. The use of colors and light produce texture and gestures that transmit what the characters are subjected to. The perception given by shot 1 and shot 2 is that of being dragged from the real world a whole different world disconnected from what you know to be normal.

Animation:

In the early 2000's the animation was produced by humans, and then with the help of expensive supercomputers this shots, such as avenue amy, were to give the actor the look of animated characters. Now, in the late 2004's animation had incredibly improved to a high level. Shots 1 and 2 gave animation a whole different meaning, allowing objects to be dimensionally created in a fast and more simple way than how it previously was with standard algorithm. Shot 1 shows improves in the object motions and speed, and shot 2 demonstrated how objects can be dimensionally created in a way they can be sensitive to any other object visual implementation. Finally, in late 2011's shot 1 shows the big improvement in animation. Animated figures look just as real as human, and their posture, gestures, and motion simulations look very natural, just like if it were a different real dimension. Shot 2 shows a better improvement in animated figures emotions and expressions are simulated through computer. Most of the tale of expressed and simulated by face expressions and body motions combined with a very good technique of lighting and object shading.

Architecture:

In 2000's the architecture was just taking more place in the world of computer, I found shot avenue emy to do a good justice to it since the architecture was not really as good as other shots because of the subtle image manipulation type that is was being used. However, in late 2004's was already part of the computer graphics world. Shot 1 demonstrate the improvement when trying to simulate a war environment. The texture and pixel shading, along with the animation techniques made the architecture of this shot to look virtually real, since it was more gaming based looking. Shot 2 was more dimensionally objective, which helps the architecture to improve in terms of making it look more real and interactive with its characters. Finally, in late 2011's a whole new world was born. Architecture was completely transformed because texture, lighting, pixel shading, animation and other visual effects advanced very quickly. A good proof to this is shot 1; the graphic simulation of this architecture makes you feel like if you are being dragged to a different dimension. Shot 2 also shows how virtual gaming world has advanced from gaming based to a real and virtual world mixture.

Overall Advancement

All these changes from the early 2000's to late 2011 have given computer graphics a huge turn in our society. It passed from being a simple way of displaying objects, to the construction of a whole different world that can easily help us to shape and improve ours in a very inexpressive and efficient way.